

Timing-Dependent Spiking Neural Network: Board-Level Hardware Implementation with Photoelectroactive Van der Waals Synapses

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The rapid growth of unstructured data in applications such as autonomous systems and edge AI underscores the urgent need for energy-efficient, real-time computing exemplified by biological brains, where synaptic weights are adjusted according to the timing of neural spikes, known as spike-timing-dependent plasticity (STDP). This work presents the first experimental realization of a multi-channel timing-dependent spiking neural network (TD-SNN) at the board-level by integrating photoelectroactive synaptic devices with an analog leaky integrate-and-fire (LIF) neuron circuit. The synaptic devices exploit the precise timing dependency between electrical presynaptic and optical postsynaptic spikes to emulate STDP, enabling reversible and bidirectional modulation of synaptic weights through photoelectroactive doping. By engineering the shape of presynaptic pulses, the devices demonstrate diverse biological STDP learning rules, including Hebbian, anti-Hebbian, all-LTP, and all-LTD. Integrated single- and multi-channel networks exhibit self-learning, system-level adaptive, and competitive behaviors. Experimentally extracted STDP parameters are implemented in SNN simulations, where network performance is determined by the long-term potentiation/depression area ratio (LTP/D area ratio, *PDR*) of the STDP curve. When $PDR \geq 1.25$, robust pattern classification is achieved, reaching up to 90.9% accuracy on MNIST tasks. These results mark a milestone in timing-dependent neuromorphic hardware, demonstrating device-level feasibility toward adaptive and real-time learning hardware.

1. Introduction

Biological neural networks perform computation and storage simultaneously through an event-driven parallel architecture that selectively activates neurons and synapses, enabling energy-efficient operation.^[1–5] Inspired by this principle, spiking neural networks (SNNs) have gained prominence as a powerful computing paradigm capable of real-time, event-driven, and spatiotemporal processing to meet the increasing demand for unstructured data processing.^[2,6–8] In general, hardware-based SNNs can be categorized into timing-independent SNNs (TID-SNNs) and timing-dependent SNNs (TD-SNNs), depending on whether spike timing information is utilized during network learning.^[9,10] TID-SNNs operate in a two-step process of read (inference) and write (train), where the output is calculated based on real-valued input signals, and then synaptic weights are globally updated (Figure 1a,c).^[8,11,12] This structure enables the application of feedforward and error backpropagation algorithms from conventional artificial neural networks (ANNs) with only a modest

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adjustment to accommodate spike-driven processing, making TID-SNNs the primary focus of early research due to their ease of implementation.^[13,14] In TID-SNNs, however, all neurons and synapses are synchronously activated, resulting in repeated redundant computation and memory access while failing to reflect temporal information. As a result, recent research has increasingly turned to TD-SNNs, in which the timing difference (Δt) between the presynaptic and postsynaptic spikes arriving at the synapse is exploited so that read (inference) and write (weight update) are performed simultaneously (in a single step) through continuous neural activity (Figure 1b,d, and Table 1).^[10,15–18]

In the early stage of TD-SNN research, CMOS-based TD-SNN systems such as TrueNorth,^[19] Loihi,^[20] and Tianjic^[21] demonstrated high energy efficiency but required complex circuitry with at least 18 transistors per neuron and 6 transistors per synapse.^[22] To address these issues, instead of the conventional three-terminal structure where a single gate controls the device conductance (synaptic weight), a four-terminal dual-gate field-effect transistor (FET) architecture has been proposed.^[23,24] In this architecture, the two gate terminals are used as the presynaptic and postsynaptic spike input ports so that the Δt between the signals applied to the gates is directly reflected in the device operation. Specifically, when voltages are applied to the top and bottom gates at different times, the resulting variation in the internal electric field modulates the synaptic weight, thereby allowing the device to emulate the temporal dynamics of biological synapses.^[25] However, this approach increases device complexity, fabrication difficulty, and power consumption, so further research is needed for large-scale integration and practical applications.^[26]

In addition to such silicon-based approaches, research on artificial synapses using various new materials, including organic,^[27,28] oxide,^[29,30] and 2D^[31–35] semiconductors, has been actively pursued. Among them, atomically thin 2D materials, such as graphene, molybdenum disulfide (MoS_2), tungsten disulfide (WS_2), tungsten diselenide (WSe_2), hexagonal boron nitride ($h\text{-BN}$), and black phosphorus (BP), exhibit almost no surface defects, thereby providing stable interfaces, where defect formation and oxidation are effectively suppressed, even when stacked as heterostructures. This structural advantage is highly suitable for forming van der Waals (vdW) heterostructures and enables the implementation of tailored electronic structures.^[36–38] By precisely engineering the band alignment through a combination of diverse 2D materials, such vdW heterojunctions can effectively fulfill the complex functional requirements of intelligent semiconductor devices, such as artificial

synapses, including synaptic weight modulation, spike-timing-dependent plasticity (STDP) behavior, and multimodal signal responsiveness.^[36,39] To date, research on 2D semiconductor-based artificial synapses has mainly centered on implementing STDP rules at the single-device level or on evaluating device characteristics. As a result, there is a lack of comprehensive studies addressing self-learning, multi-channel operations, and system performance evaluation at the network level using experimental data.^[40,41]

Here, we present the first printed circuit board (PCB)-level realization of a multi-channel TD-SNN system by integrating photoelectroactive (PE) artificial synaptic devices based on 2D semiconductors with analog leaky integrate-and-fire (LIF) neuron circuits. In this system, the artificial synapse can emulate the bidirectional and reversible STDP of biological synapses in hardware, depending on the timing difference between the electrical presynaptic spike (gate voltage, $V_{\text{Pre,G}}$) and the optical postsynaptic spike (ultraviolet light, V_{Post}). When $V_{\text{Pre,G}}$ and V_{Post} overlap within a specific timing window ($\Delta t > 0$ or $\Delta t < 0$), ultraviolet (UV) illumination triggers ionization or neutralization of inherent defects in $h\text{-BN}$, and simultaneously, the gate voltage makes the generated electrons or holes migrate into the MoS_2 channel, thereby modulating device conductance to increase (long-term potentiation, LTP) or decrease (long-term depression, LTD). Pulse engineering further enables the selective realization of diverse biological learning rules, such as Hebbian, anti-Hebbian, all-LTP, and all-LTD. Notably, by implementing single-channel and multi-channel synaptic networks, the hardware system can realize self-learning, competitive and cooperative mechanisms, and multi-channel feedback operations that mimic brain-like functionality. Moreover, by linking the STDP characteristics extracted from the PE synapses to SNN simulations, we validate self-learning and hardware-based SNN performance integrated across the device, network, and system levels. This integrated platform enables real-time adaptive learning by autonomously modulating synaptic weights according to spike timing differences, suggesting the potential for next-generation low-power, high-performance neuromorphic computing technologies.

2. Results and Discussion

Figure 2a shows a conceptual illustration of a synapse, a neuron, and a spiking neural network in a biological brain, along with their corresponding artificial counterparts: a PE synapse, a LIF neuron circuit, and a TD-SNN. In the biological synapse on the left side (blue dashed box), when an action potential (a presynaptic spike) occurs in the presynaptic neuron, it travels down the axon terminal. Then, it triggers the release of neurotransmitters into the synaptic cleft (a synapse). The released neurotransmitters bind with receptors on the postsynaptic dendrite, transmitting the signal to the cell body of the postsynaptic neuron, which induces a postsynaptic spike. Here, the magnitude of the transmitted signal via a synapse varies depending on the strength of the synaptic connection, referred to as the synaptic weight. This weight is modulated by the Δt between the arrival of the presynaptic and postsynaptic spikes at the synapse. When the presynaptic spike arrives at the synaptic cleft before the postsynaptic spike ($\Delta t > 0$), it indicates that the presynaptic spike stimulus

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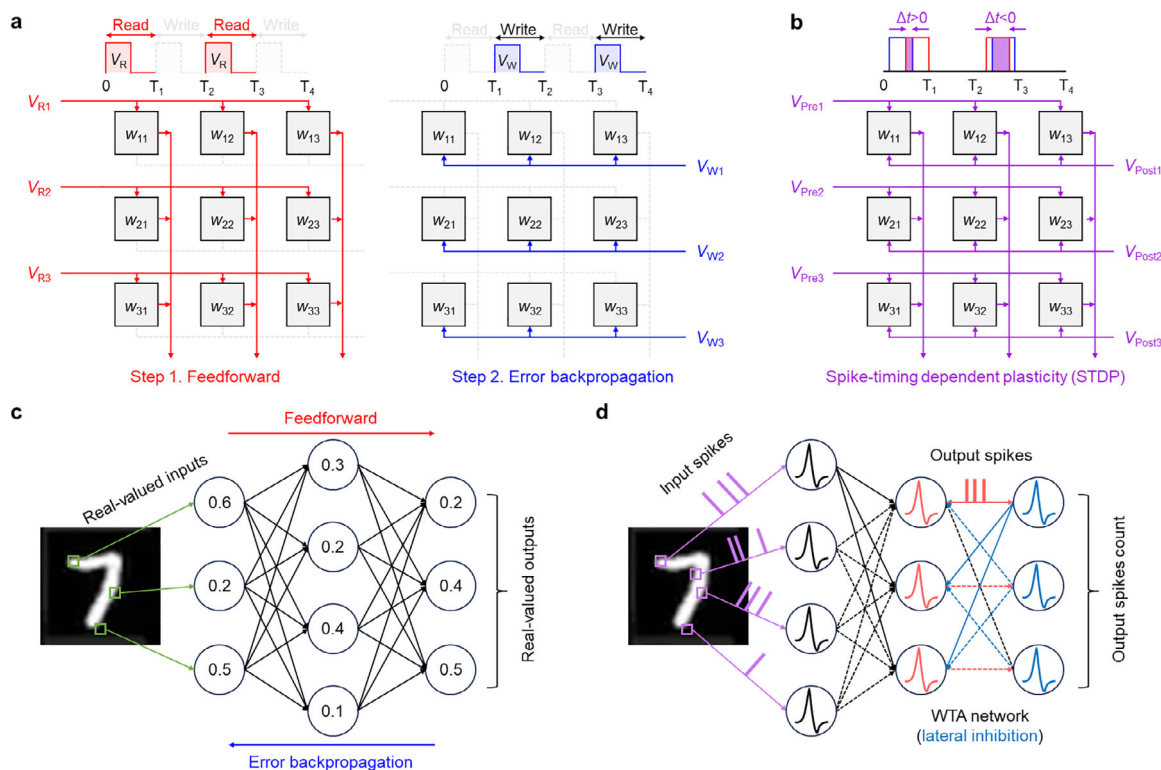


Figure 1. Schematic illustration of learning and computation schemes in TID-SNN and TD-SNN. a) Two-step process in TID-SNN showing sequential feedforward (inference) followed by backpropagation (train). b) Single-step mechanism in TD-SNN where read (inference) and write (weight update) occur simultaneously at each synapse via STDP. c) Synchronous network-wide activation in TID-SNN, where neurons and synapses are co-activated during inference and training. d) Event-driven computation in TD-SNN with synapse activation co-related to receiving spikes for efficient and real-time operation.

has greatly contributed to triggering the firing of the postsynaptic neuron. Such causal timing tightly binds the two firing events and lays the foundation for long-term memory, thereby strengthening the synaptic connection (i.e., the weight), a process referred to as LTP. In contrast, when the postsynaptic spike occurs first ($\Delta t < 0$), the causality between the firings is broken, resulting in the synaptic weight gradually weakening, a process called LTD. This biological principle of synaptic weight modulation is known as

Table 1. Comparison of the key characteristics of TID-SNN and TD-SNN. In TID-SNNs, the entire network is synchronously activated for each computation, requiring repeated global updates via error backpropagation and inevitably resulting in high energy consumption. In contrast, TD-SNNs process information through discrete spikes with synaptic updates occurring locally only when the relevant neuron fires. This event-driven architecture minimizes unnecessary computations and greatly improves energy efficiency, positioning TD-SNNs as optimal candidates for ultra-low-power neuromorphic systems.

Key characteristics	TID-SNN	TD-SNN
Learning process	Two-step	Single-step
Signal propagation	Synchronous	Event-driven
Learning algorithm	Feedforward & Error backpropagation	STDP rules
Energy efficiency	Low	High

STDP and represents one of the fundamental operating mechanisms of biological synapses.^[42] To mimic this behavior, we implemented a transistor-type artificial synapse using a heterojunction structure of MoS_2 and $h\text{-BN}$, where the channel conductance is modulated according to the timing difference between the voltage and light spikes. This device operation originates from the photoelectroactive doping (PE-doping) effect observed in a 2D vdW heterojunction structure.^[43]

In our system, the presynaptic spike consists of two distinct voltage pulses applied to the gate ($V_{\text{Pre,G}}$) and drain ($V_{\text{Pre,D}}$). The postsynaptic spike is delivered as an UV light-emitting diode (LED) light pulse signal activated by postsynaptic voltage (V_{Post} ; see Figure S1, Supporting Information). Figures S2–S6 (Supporting Information) show a detailed mechanism of the conductance modulation in the artificial synapse. When a light pulse is applied under a negative gate voltage ($V_{\text{Pre,G}} < 0$), the photogenerated electrons from the inherent defects in the $h\text{-BN}$ layer migrate to the MoS_2 channel bypassing the positively charged defects. This change leads to an increase in the channel conductance. Conversely, under a positive gate voltage ($V_{\text{Pre,G}} > 0$), a light pulse generates electron-hole pairs from the valence band of the $h\text{-BN}$. The photogenerated electrons neutralize the positive fixed charges in the $h\text{-BN}$ layer, while the photogenerated holes migrate to the MoS_2 channel and recombine with the excess electrons, resulting in a conductance decrease. Figures S7 and S8 (Supporting Information) shows a non-volatile change in the conductivity of the

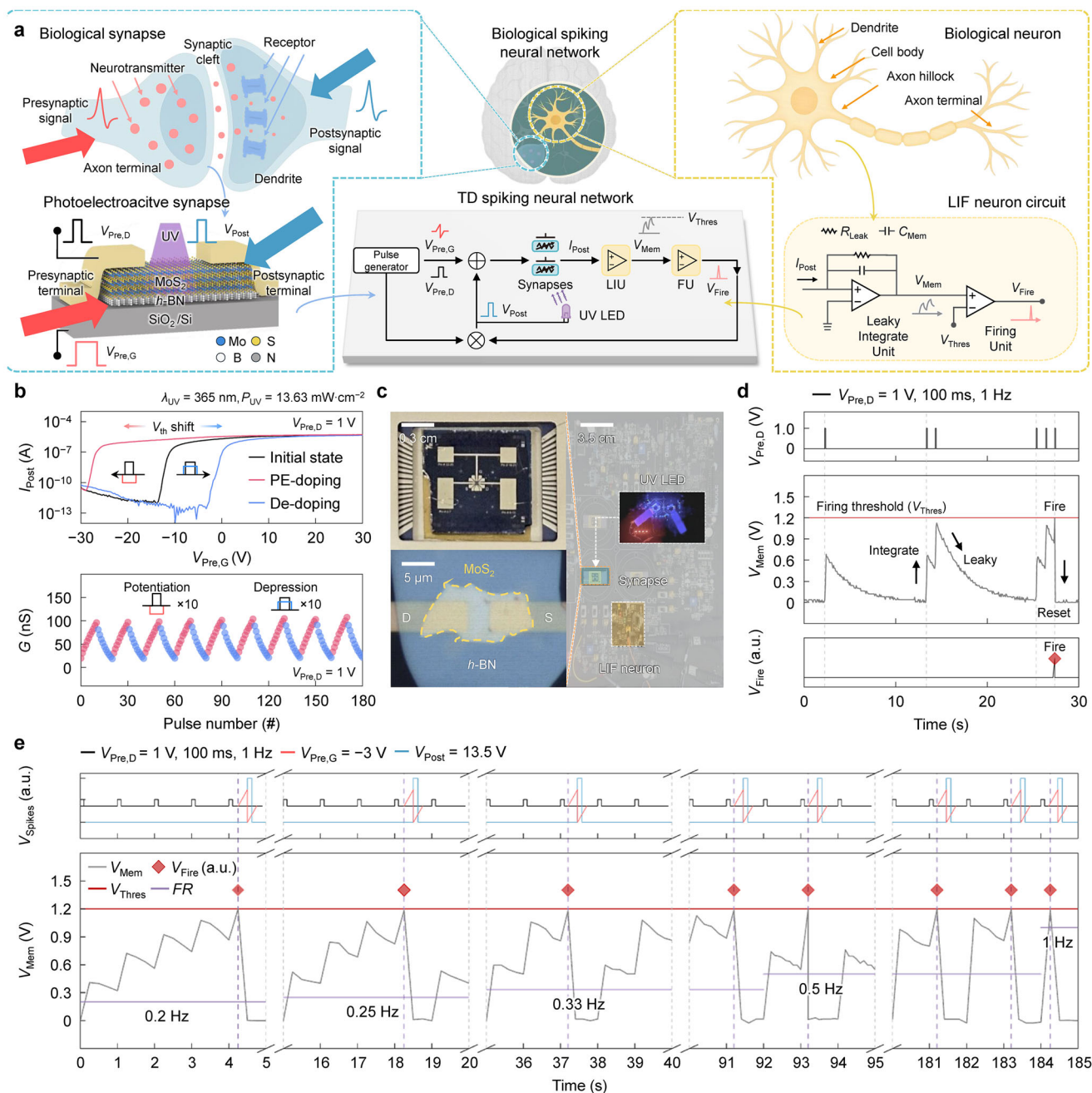


Figure 2. Configuration and operation of a TD-SNN integrating PE MoS₂/h-BN synaptic devices with a LIF neuron circuit. a) Schematic representations of biological synapse, neuron, and network, and their artificial counterparts: PE synapse, LIF neuron circuit, and TD-SNN. b) PE-doping effect in the synaptic device, showing V_{th} shift characteristics (top) and reversible conductance modulation during repeated potentiation and depression cycles (bottom). c) Photographs and OM images of the implemented TD-SNN, showing a wire-bonded synaptic device and MoS₂/h-BN heterostructure (left), integrated network and neuron circuits (right), and network during UV illumination (inset). d) Basic operation of the LIF neuron circuit, illustrating integration and exponential decay of V_{Mem} with firing triggered when V_{Mem} reaches V_{Thres} . e) Self-learning behavior of TD-SNN, demonstrating progressive synaptic potentiation through repeated stimulation, accelerating FR from 0.2 to 1 Hz.

implemented synaptic device. The photogenerated charge carriers become spatially separated at the h-BN/MoS₂ interface, and the large band offset of the h-BN blocks their recombination. The trapped charges act like a floating gate, creating an internal electric field that continuously modulates the channel conductance without further stimuli.

On the right side (yellow dashed box), a biological neuron and its functional counterpart implemented as an analog LIF neuron circuit are presented. A biological neuron consists of dendrites, a cell body, and axons, each of which is responsible for receiving, integrating, and transmitting the incoming signals, respectively.^[44] Similarly, the implemented LIF neuron circuit is

composed of a leaky integrate unit (LIU) and a firing unit (FU), which mimics the operational mechanism of biological neurons. The LIU converts the postsynaptic current (I_{Post}) generated from artificial synapses into a voltage signal through a leakage resistance (R_{Leak}), and simultaneously, the voltage signal is gradually accumulated across the membrane capacitance (C_{Mem}). This growth leads to a progressive increase in the membrane potential (V_{Mem}), which naturally decays over time due to the presence of R_{Leak} , emulating the leaky behavior observed in biological neurons. The FU compares V_{Mem} with the firing threshold (V_{Thres}) of 1.2 V. Once V_{Mem} reaches the threshold, the FU triggers a firing signal (V_{Fire}). After firing, V_{Mem} is reset to its initial state through an RC decay process. The detailed operational behavior of both the LIU and FU circuits is illustrated in Figures S9 and S10 (Supporting Information).

The center illustrations compare a biological spiking neural network in a human brain to an implemented TD-SNN through the integration of the PE synapses and the LIF neuron circuits. In the biological network, numerous neurons are interconnected through synapses, and the synaptic strength is automatically modulated according to the relative timing between the presynaptic and postsynaptic spikes. This dynamic plasticity (STDP) serves as the fundamental mechanism enabling efficient learning and memory formation in the brain. Similarly, in the implemented TD-SNN, the synaptic devices are directly integrated onto and connected with the LIF neuron circuits. The signal transmission and learning process of TD-SNN proceed as follows. First, when a V_{Pre} signal set, composed of $V_{\text{Pre,D}}$ and $V_{\text{Pre,G}}$, is applied to a synaptic device, I_{Post} is generated and delivered to the LIF neuron circuit. Then, V_{Mem} gradually increases by I_{Post} pulse train in LIU. When V_{Mem} reaches V_{Thres} , the FU generates V_{Fire} , triggering a pulse generator to produce V_{Post} , which subsequently turns on the UV LED to apply a light spike signal to the synaptic device. Through this closed feedback loop, the channel conductance of the synaptic device is automatically modulated depending on the timing difference between the input voltage spike ($V_{\text{Pre,G}}$) and the output light spike (V_{Post}), precisely following the STDP learning rule. As a result, this induces progressive strengthening or weakening of the synaptic connections based on the input patterns, ultimately enabling self-learning behavior in the TD-SNN (see Figure S11, Supporting Information).

The top panel of Figure 2b shows the $I_{\text{Post}}-V_{\text{Pre,G}}$ characteristic curves in response to the PE-doping effect. $V_{\text{Pre,D}}$ was set to 1 V, and the power of the 365 nm UV LED was $13.63 \text{ mW}\cdot\text{cm}^{-2}$ (see Figure S12, Supporting Information). Under a gate voltage of -30 V , the UV illumination for 60 s caused the threshold voltage (V_{th}) of the device to shift from -12.6 to -27.9 V . In contrast, when the gate voltage was set to 30 V , the UV LED exposure for 60 s led to a V_{th} shift from -12.6 to -1.2 V . This permanent V_{th} modulation indicates a non-volatile conductance change in the MoS_2 channel, highlighting that the device exhibits the long-term plasticity of a biological synapse.

The bottom panel of Figure 2b presents the conductance response to a pair of voltage and light spike pulses over 9 cycles, where each cycle consists of 10 potentiation ($V_{\text{Pre,G}} = -3 \text{ V}$) and 10 depression ($V_{\text{Pre,G}} = 7 \text{ V}$) pulse pairs. The pulse width was 100 ms, and the frequency was 1 Hz. When 10 potentiation pulse pairs were applied to the device, the conductance increased gradually from 19.9 to 101.2 nS. Conversely, applying 10 depression

pairs reduced the conductance, bringing it back to the initial level. The average dynamic range of the conductance ($G_{\text{Max}}/G_{\text{Min}}$) was 5.14 over 9 cycles, and its relative standard deviation (RSD) was maintained only at 1.7%, emphasizing the reliability and repeatability of the device's operation (See Figure S13 for additional endurance characteristics of synaptic device, Supporting Information). This gradual conductance modulation characteristic is a key feature of biological synapses known as long-term potentiation and depression (LTP/D).^[45]

Figure 2c presents photographs and optical microscope (OM) images of the implemented TD-SNN. The left panel shows a wire-bonded synapse chip (top) and a PE synaptic device (bottom), which consists of a MoS_2 and h -BN heterojunction structure. The right panel displays the integrated network of synaptic devices and the LIF neuron circuits. The inset image shows a photograph of the network under UV illumination corresponding to the firing event.

Figure 2d illustrates the basic operation of the LIF neuron connected with a synaptic device in the TD-SNN. When a single $V_{\text{Pre,D}}$ spike of 1 V and 100 ms was applied to the synaptic device, V_{Mem} abruptly increased from 0 to 0.7 V (Integrate) and then decayed back to the initial level (Leaky). No firing event occurred because V_{Mem} did not exceed the V_{Thres} . Next, two $V_{\text{Pre,D}}$ spikes of 1 s interval caused V_{Mem} to increase to 1.1 V, but this value still did not reach V_{Thres} . Lastly, after three consecutive $V_{\text{Pre,D}}$ spikes, V_{Mem} exceeded the V_{Thres} , and consequently, the FU generated V_{Fire} . The V_{Mem} was then immediately reset to the initial point to prepare for the next firing event. This result confirms that the proposed TD-SNN successfully mimics the LIF behavior of biological neural networks in response to external stimuli.

Figure 2e demonstrates the self-learning operation of the TD-SNN. During the initial phase (0–5 s), the LIF neuron required five $V_{\text{Pre,D}}$ pulses of 1 Hz to reach the V_{Thres} , resulting in a firing rate (FR) of 0.2 Hz. The firing event activated V_{Post} to turn on the UV LED. Because $V_{\text{Pre,G}}$ had a negative amplitude during this phase, the conductance of the synaptic device increased, leading to the LTP phenomenon. As the conductance increased by firing events, V_{Mem} was accumulated more rapidly in response to each subsequent $V_{\text{Pre,D}}$ pulse, thereby reducing the number of spikes required for the firing event. In the graph from 15 to 20 s, the four pulses triggered firing, where the FR was 0.25 Hz. As a result of such a positive feedback loop, the FR was accelerated from 0.2 to 1 Hz (five-fold) for 185 s. These results show that the implemented TD-SNN successfully emulated the potentiation behavior commonly observed in biological synapses.^[42]

Beyond the classic Hebbian rule—where LTP occurs for $\Delta t > 0$ and LTD for $\Delta t < 0$ —various STDP learning rules have been observed in biological systems.^[46,47] To replicate this diversity, we engineered the shape of the $V_{\text{Pre,G}}$ pulse to precisely control the conductance change (ΔG) as a function of Δt , thereby enabling the implementation of multiple types of STDP curves. Figure 3a shows a schematic illustration of the custom-designed circuit used to precisely control the Δt between $V_{\text{Pre,G}}$ and V_{Post} , thereby facilitating the implementation of STDP in the PE synaptic device. In this circuit, a semiconductor parameter analyzer (4200A-SCS, Keithley) controls the Δt for various $V_{\text{Pre,G}}$ pulse shapes. $V_{\text{Pre,G}}$ is connected to the gate terminal of the synaptic device, while V_{Post} is connected to that of a commercial MOS-FET (2N7000, Onsemi) that activates the UV LED. The $V_{\text{Pre,D}}$ is

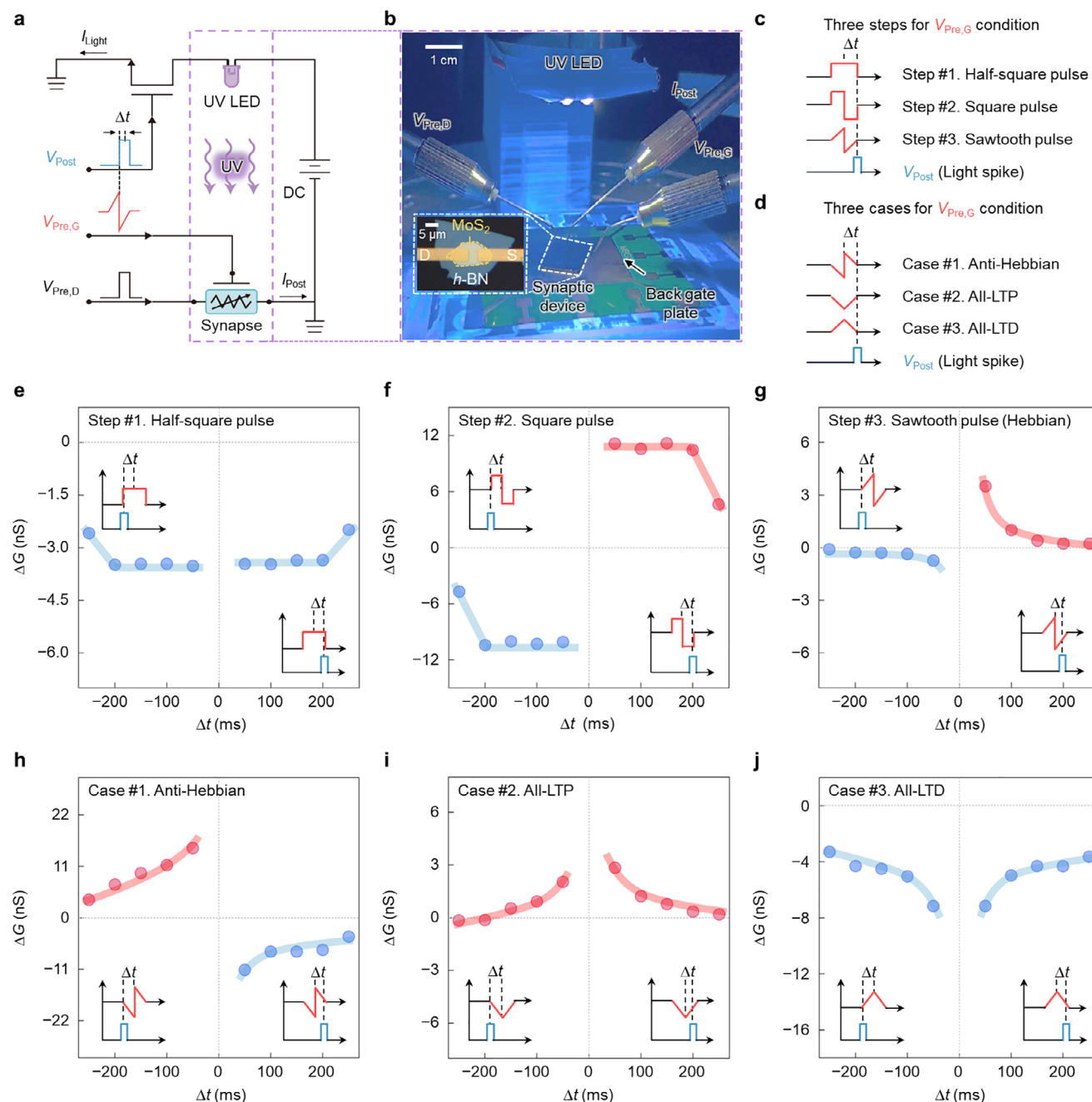


Figure 3. Pulse shape engineering to realize Hebbian and diverse STDP learning rules. a) Schematic of the custom-designed circuit used to precisely control the Δt between $V_{Pre,G}$ and V_{Post} . b) Photograph of the measurement setup for the PE synapse and OM image of the $\text{MoS}_2/\text{h-BN}$ heterojunction synapse (inset). c) Three-step $V_{Pre,G}$ pulse engineering for the Hebbian learning rule: half-square (step #1), square (step #2), and sawtooth pulses. d) Three additional $V_{Pre,G}$ pulse cases for non-Hebbian STDP characteristics: inverted sawtooth (anti-Hebbian), negative triangular (all-LTP), and positive triangular (all-LTD). e–g) Hebbian STDP learning rules obtained from Step #1–#3: e) Half-square pulse (LTD only). f) Square pulse ($\Delta t > 0$: LTP, $\Delta t < 0$: LTD). g) Sawtooth pulse (Hebbian). h–j) Non-Hebbian STDP curves achieved by Case #1–#3: h) Inverse sawtooth pulse (anti-Hebbian). i) Negative triangle pulse (all-LTP). j) Positive triangle pulse (all-LTD).

connected to the drain terminal to induce the I_{Post} , thereby reading the device's channel conductance (G). Figure 3b presents a photo of the measurement setup for the PE synaptic device integrated with the timing control circuit. The inset image shows an OM image of the $\text{MoS}_2/\text{h-BN}$ heterojunction structure, and

the channel of the synaptic device has a width/length (W/L) ratio of $5/4 \mu\text{m}$. The UV LED is positioned above the synaptic device, while the $V_{Pre,G}$ is applied to the gate through the back gate plate. Using this measurement setup, we conducted experiments to replicate not only the Hebbian learning rule (Figure 3c,e–g) but

also other diverse STDP characteristics (Figure 3d,h–j) by modulating the shape of the $V_{\text{Pre,G}}$ pulses. Figure 3c displays three pulse engineering steps: Half-square, square, and sawtooth to achieve the well-known Hebbian STDP learning rule. In Step #1 (Figure 3e), the half-square $V_{\text{Pre,G}}$ pulse maintained a consistent positive electric field at any Δt with V_{Post} , resulting in only LTD being observed for both $\Delta t > 0$ and $\Delta t < 0$ (Figure 3e). When Δt was ± 250 ms, the magnitude of ΔG decreased because V_{Post} partially overlapped with $V_{\text{Pre,G}}$ (see Figure S14, Supporting Information). To achieve both the LTP and LTD characteristics, a square pulse with alternating negative and positive electric fields was applied as $V_{\text{Pre,G}}$ (Step #2). This pulse scheme successfully generated LTP at $\Delta t > 0$ and LTD at $\Delta t < 0$ (Figure 3f). To address the absence of ΔG decay with increasing Δt , which is typically observed in Hebbian STDP curves, a sawtooth pulse was subsequently introduced as $V_{\text{Pre,G}}$ (Step #3). The gradual modulation of the electric field facilitated the successful implementation of both gradual LTP and LTD characteristic curves, which exhibited a decay progressively with increasing Δt (Figure 3g). These experiments demonstrated that Hebbian STDP learning can be effectively achieved via accurate pulse engineering.

Meanwhile, biological synapses also exhibit diverse STDP characteristics beyond the Hebbian learning rule, such as anti-Hebbian, all-LTP, and all-LTD. To mimic such learning rules, three $V_{\text{Pre,G}}$ pulse cases (Cases #1–#3) were introduced, as illustrated in Figure 3d. In Case #1, a sawtooth pulse inverted along the horizontal axis was applied, resulting in successful realization of the anti-Hebbian STDP curves (Figure 3h). Anti-Hebbian STDP is typically observed in the cerebellum, where synaptic inhibition is crucial for learning and adaptation.^[47,48] Figure 3i displays the obtained all-LTP STDP curves, where only LTP occurs for both $\Delta t > 0$ and $\Delta t < 0$. The negative triangular pulse caused the synaptic weight of the device to be strengthened exclusively regardless of Δt . This all-LTP behavior has been observed at CA3–CA3 recurrent synapses in the hippocampus, a region associated with learning and memory.^[49] Figure 3j illustrates the all-LTD STDP curves, which are vertically inverted relative to Case #2. The positive triangular pulse resulted in the suppression of synaptic weight at any Δt . All-LTD behavior can be induced in the visual cortex—a region critical for sensory processing and adaptive plasticity—when low levels of norepinephrine selectively activate α_1 -adrenergic receptors.^[50,51] These results demonstrate the effectiveness of pulse engineering in replicating various biological STDP curves, providing insights into the learning and adaptation mechanisms of biological synapses.

To evaluate the self-learning characteristics of the implemented TD-SNN, three distinct cases of a biological neural network were constructed as follows: Case #1, positive feedback; Case #2, negative feedback; and Case #3, asymmetric regulation. Figure 4a presents a simplified schematic of a neural network with a single-channel positive feedback loop and a corresponding Hebbian pulse scheme. The circuit consists of a single artificial synapse (Channel 1), an LIF circuit, and a UV LED. In this single-channel network, postsynaptic spikes are always triggered by presynaptic spikes, ensuring that Δt retains a positive value. To induce 0a Hebbian response under the $\Delta t > 0$ condition, $V_{\text{Pre,G,1}}$ was applied in the form of a sawtooth pulse. The network operates as follows: I_{Post} generated by $V_{\text{Pre,D,1}}$ charges C_{Mem} of the LIF circuit, resulting in an increase of V_{Mem} . When V_{Mem} reaches

V_{Thres} , V_{Post} is generated, causing the UV LED to be activated. The negative segment of the sawtooth pulse coincides with the timing of the V_{Post} activation, resulting in the induction of LTP in the artificial synapse according to the Hebbian learning rule. Figure 4b shows the operational results of the corresponding neural network. When the $V_{\text{Pre,1}}$ pulse signals ($V_{\text{Pre,D,1}} = 1$ V, $V_{\text{Pre,G,1}} = -7$ V) with a frequency of 1 Hz were applied to the synaptic device, V_{Mem} increased in a stepwise manner, and firing occurred when V_{Mem} reached V_{Thres} approximately at 4 s. At this point, according to the Hebbian pulse scheme, LTP was induced and the G increased from 8.6 to 8.7 μS . This increase in conductance led to an increase in I_{Post} , causing V_{Mem} to reach V_{Thres} more rapidly, which in turn results in a faster firing event and forms a positive feedback loop. As the $V_{\text{Pre,1}}$ pulses were continuously applied, the G further increased from 8.6 to 9.2 μS over 35 s, and the number of input pulses required to trigger a firing event decreased from 4 to 1. Such a conductance increase driven by positive feedback effectively emulates the LTP-driven self-learning mechanism that has been observed in biological neural networks.^[52]

Figure 4c illustrates a single-channel negative feedback neural network connected with a single artificial synapse in Channel 2 and the corresponding anti-Hebbian pulse scheme. Owing to the single artificial synapse configuration, the presynaptic spike always precedes the postsynaptic spike, so Δt is fixed as a positive value similar to Case #1. Upon applying a reverse sawtooth pulse to $V_{\text{Pre,G,2}}$, the timing of V_{Post} consistently overlaps with the positive portion of $V_{\text{Pre,G,2}}$, which continuously induces LTD in the synapse. In Figure 4d, as $V_{\text{Pre,2}}$ pulses ($V_{\text{Pre,D,2}} = 1$ V, $V_{\text{Pre,G,2}} = 40$ V) of 1 Hz were applied to the synaptic device, V_{Mem} reached V_{Thres} in ≈ 0.2 s. This caused a firing event, and LTD was induced, subsequently decreasing the G from 9.0 to 8.9 μS . With continued stimulation, the conductance gradually decreased from 9.0 to 7.9 μS over ≈ 130 s. As the conductance decreased, the increase of V_{Mem} produced by each input pulse became smaller. Therefore, a greater number of $V_{\text{Pre,2}}$ pulses were required to accumulate a sufficient charge to reach V_{Thres} and trigger subsequent firing. As a result, the number of pulses required for firing increased from 1 to 4, and simultaneously, the interval between the firing events was extended. This negative feedback loop effectively suppresses excessive excitation by reducing synaptic strength, thereby emulating homeostatic plasticity at the hardware level.^[53]

As described in the previous section, a single synapse neural network should only have either LTP or LTD individually because Δt is always positive. However, previous studies have been limited to the implementation of neural systems by connecting only a single synaptic channel to neuron circuits, typically focusing on investigating individual forms of synaptic plasticity (see Table 2). However, in biological systems, the firing of a neuron is determined by the combined weights of multiple synapses, and thereby both LTP and LTD occur simultaneously.^[54] To implement these biological features at the hardware level, we designed an asymmetric feedback network that integrates a potentiation-inducing channel (Ch. 1) and a depression-inducing channel (Ch. 2) in parallel.

Figure 4e illustrates the network architecture and the pulse schemes applied to each channel. In this network, the $I_{\text{Post,1}}$ and $I_{\text{Post,2}}$ generated by each channel were delivered to the LIF circuit, where V_{Mem} was integrated, and firing was triggered upon reaching V_{Thres} . Figure 4f and Video S1 (Supporting Information)

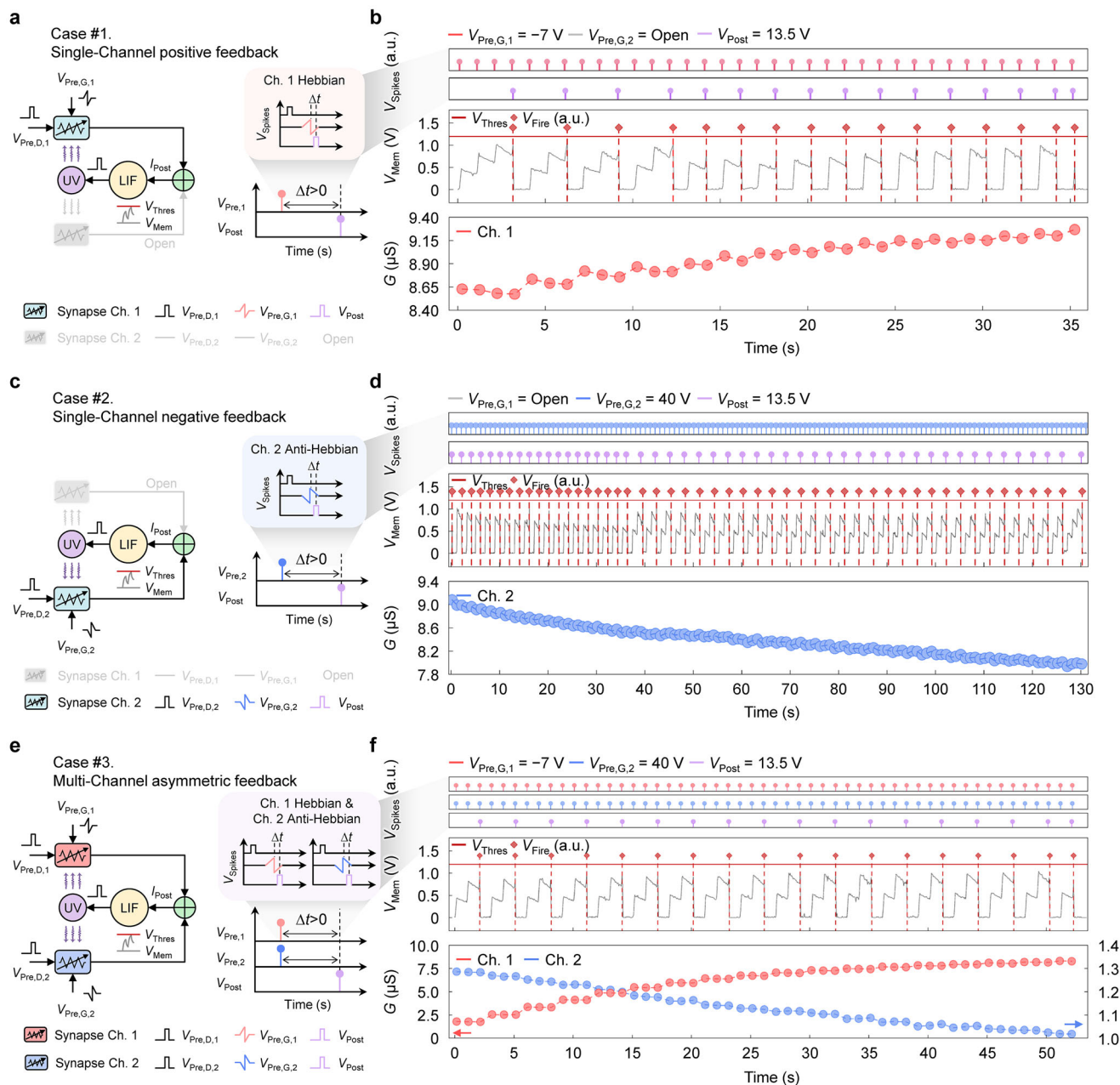


Figure 4. Feedback-driven self-learning characteristics in TD-SNN with single- and multi-channel synapses. a, c, and e) Simplified schematic of the TD-SNN configuration: a) Single-channel positive feedback network (Ch. 1) with a Hebbian pulse scheme; c) Single-channel negative feedback network (Ch. 2) with an anti-Hebbian pulse scheme; and e) Multi-channel asymmetric network (Ch. 1 and 2) with channel-specific pulse waveforms applied in parallel. b, d, and f) Operational characteristics of each network: b) Self-reinforcing conductance increase and accelerated firing in the single-channel positive feedback, d) Progressive conductance decrease and suppressed firing in the single-channel negative feedback, and f) Adaptive self-learning with competitive and cooperative modulation of synaptic weights in the multi-channel asymmetric feedback configuration.

show the operational characteristics of the network. When $V_{Pre,1,2}$ pulses ($V_{Pre,D} = 1$ V, $V_{Pre,G,1} = -7$ V, $V_{Pre,G,2} = 40$ V) were applied at 1 Hz, the first firing occurred at ≈ 2 s. At this point, the conductance for Ch. 1 increased from 1.78 to 2.55 μS , while conductance for Ch. 2 decreased from 1.28 to 1.26 μS . These opposing conductance modulations resulted from the channel-specific presynaptic waveforms: a sawtooth pulse inducing LTP in Ch. 1 and a reversed sawtooth pulse causing LTD in Ch. 2. With continued

application of the $V_{Pre,1,2}$ pulses, this contrast in conductance became more pronounced. After 53 s, the conductance of Ch. 1 increased from 1.78 to 8.28 μS , while that of Ch. 2 decreased from 1.28 to 1.01 μS . Although the two channels exhibited distinct forms of synaptic plasticity, both contributed simultaneously to the synaptic dynamics of the neural circuit. This interaction regulated the charging rate of C_{Mem} , thereby reducing the number of presynaptic pulses required to reach V_{Thres} from three to

Table 2. Benchmark of SNN implementations integrating artificial synaptic devices with neuron circuits. Most previous works have either simulated the neuron system or connected a single synaptic channel to the neuron circuit. In contrast, our study demonstrates a dual-channel neural network along with integration of a pulse generator, neuron circuit, and synaptic devices on a single PCB board.

Material	Synaptic device	SNN Type	Neuron Implementation	# of Synaptic Channels	PCB-level Implementation	References
<i>h</i> -BN/MoS ₂	PE transistor ^{a)}	TD-SNN	Experiment	2	Fully	Our work
HZO/IGZO	FeFET	TID-SNN	Experiment	1	Partial	[55]
CuInP ₂ S ₆ / In ₂ Se ₃ , WSe ₂	FeFET ^{b)}	TID-SNN	Emulation	2	No	[56]
Ag/SnO ₂ /Pt	Memristor	TID-SNN	Experiment	1	Partial	[57]
CuInP ₂ S ₆ /GaN	FeHEMT	TID-SNN	Experiment	1	No	[58]
Ag/ <i>h</i> -BN/Au	Memristor	TD-SNN	Experiment	1	No	[17]
ReS ₂	Neuristor	TD-SNN	Experiment	1	No	[59]

^{a)} Photoelectroactive transistor. ^{b)} Ferroelectric FET.

two and shortening the firing interval. This adjustment in firing timing indicates the feedback-regulated capability of the implemented neural network, showing that repeated synaptic activities modulate the network's subsequent responsiveness. In particular, the multi-channel structure realized by the parallel integration of excitatory and inhibitory synaptic channels is effectively mimicking the convergent and competitive dynamics observed in biological synapses.^[60,61] Consequently, the system demonstrated autonomous weight adaptation and dynamic firing control based on accumulated inputs. These characteristics underscore a core feature of neuromorphic learning: self-regulated plasticity emerging from feedback-driven synaptic interaction.

Biological synapses exhibit diverse STDP characteristics to enable various learning and adaptation processes.^[62] Especially, the Hebbian STDP is the most important for understanding these biological mechanisms. Therefore, it is essential to experimentally implement various Hebbian STDP forms of the synaptic device to explore its mechanisms in hardware-based neural networks (HW-NNs). **Figure 5a** presents a side-view schematic of the PE synaptic device and three experimental conditions designed to examine the dependency of the Hebbian STDP curves under the following parameters: Condition #1, the conductance level at which the STDP curve was extracted; Condition #2, the presynaptic voltage amplitude applied to the gate terminal; and Condition #3, the LED current amplitude that activated the UV light signal. **Figure 5b** is a schematic representation of the expected Hebbian STDP characteristic curve. Depending on the three conditions, it is anticipated that the magnitude and symmetry of the STDP curve can be modulated.

Figure 5c–e displays the diverse STDP curves extracted from the experimental results corresponding to the three conditions outlined in **Figure 5a**. To ensure statistical reliability, the ΔG values for the STDP characteristic curves were extracted from data averaged over the 5 data points of the LTP/D curve. In Condition #1 (**Figure 5c**), the ΔG values were extracted at three distinct conductance levels, specifically Low (3 nS), Middle (20 nS), and High (100 nS). During this experiment, the peak amplitude of $V_{\text{Pre,G}}$ was fixed at ± 4.5 V and the amplitude of I_{Light} was fixed at 0.1 A. In the Low case, LTP showed a stronger response than LTD, whereas in the Middle conductance state, LTP and LTD were nearly balanced, and in the High conductance state, LTD was

more pronounced than LTP. This trend can be explained by the influence of the initial conductance state on ΔG . At a low conductance state, such as 3 nS, a further reduction in conductance is physically limited. Instead, it is easier for the conductance to increase, resulting in a more pronounced LTP response. Conversely, at a high conductance state near 100 nS, increasing conductance becomes difficult whereas decreasing conductance is more favorable, leading to a dominant LTD response while LTP is suppressed. This result shows conductance level-dependent STDP behavior, where the ΔG is highly dependent on the current conductance state within the dynamic range of the LTP/D curve.^[63] In **Figure 5d** (Condition #2), when the $V_{\text{Pre,G}}$ amplitude increased, ΔG in both the LTP and LTD regions exhibited a consistent upward trend. The peak amplitude of $V_{\text{Pre,G}}$ was varied from ± 4 to ± 5.5 V, and I_{Light} was fixed at 0.1 A. These results suggest that as the peak amplitude of $V_{\text{Pre,G}}$ increases, the corresponding electric field strengthens, leading to enhanced carrier migration and more effective conductance modulation within the synaptic channel. Consequently, the observed variations in ΔG highlight the crucial role of gate voltage strength in tuning synaptic plasticity. Condition #3 shows a similar trend as Condition #2. As the LED current amplitude increased from 0.1 to 0.3 A when $V_{\text{Pre,G}}$ was fixed at -4.5 V, the ΔG at $\Delta t = 50$ ms varied from 2.6 to 6.5 nS. The observed ΔG enhancement is attributed to the increased UV intensity at higher I_{Light} amplitudes, which facilitates stronger ionization of inherent defects within the *h*-BN layer, thereby enhancing the channel conductance modulation. These results confirm the predicted STDP variations, demonstrating that synaptic weight updates can be precisely controlled by tuning the conductance level, voltage amplitude, and optical input. Such tunability reflects the adaptive mechanisms of biological synapses,^[64] where intrinsic and extrinsic stimuli dynamically regulate synaptic strength, underscoring its potential for neuromorphic applications.

To quantitatively assess the variations in the STDP curves across Conditions #1–#3, we extracted the areas of the LTP (range of Δt from 50 to 250 ms) and LTD (range of Δt from -50 to -250 ms) regions from each curve and calculated the LTP/D area ratio (*PDR*) (see **Figure S15**, Supporting Information). In **Figure 5f**, for the Low conductance state, the LTP area was significantly larger than the LTD area. However, as the conductance

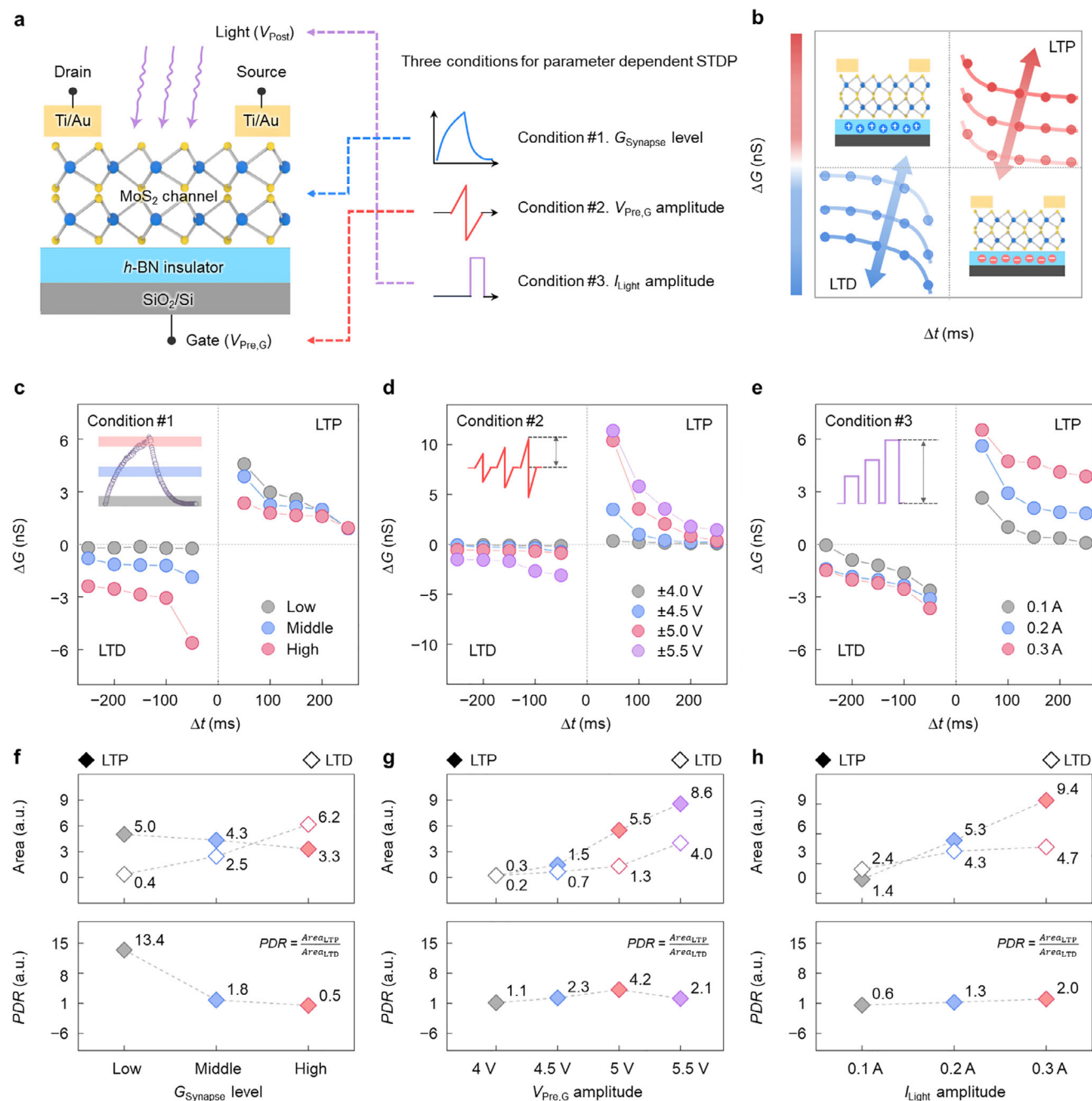


Figure 5. Tunable Hebbian STDP characteristics in PE synaptic devices under diverse experimental conditions. a) Side-view of the PE synaptic device and three experimental conditions: Condition #1, the conductance level at which the STDP curve was extracted; Condition #2, the gate voltage amplitude applied to the gate terminal; and Condition #3, the LED current amplitude that activated the UV light signal. b) Schematic representation of the expected Hebbian STDP characteristic curve. c–e) STDP curves under three conditions: c) Different initial conductance levels; d) Varying $V_{Pre,G}$ amplitudes; and e) varying I_{Light} amplitudes. f–h) Quantitative comparison of PDR for each condition: f) PDR for different conductance states; g) PDR as a function of $V_{Pre,G}$ amplitude; and h) PDR as a function of the I_{Light} amplitude.

level increased to the Middle and High states, the LTP and LTD areas gradually balanced out, and ultimately the LTD response became dominant. Accordingly, the PDR values decreased notably from 13.4 (Low) to 1.8 (Middle) and further down to 0.5 (High). These trends were similar to the characteristics observed in Condition #1. Figure 5g presents the impact of varying the $V_{Pre,G}$

from ± 4.0 to ± 5.5 V on the STDP asymmetry. As $V_{Pre,G}$ increased, both the LTP and LTD areas showed a progressive increase, consistent with the trends observed under Condition #2. This effect arises because larger gate voltages produce stronger electric fields across the synaptic channel, thereby accelerating carrier migration and enhancing conductance modulation. Consequently, the

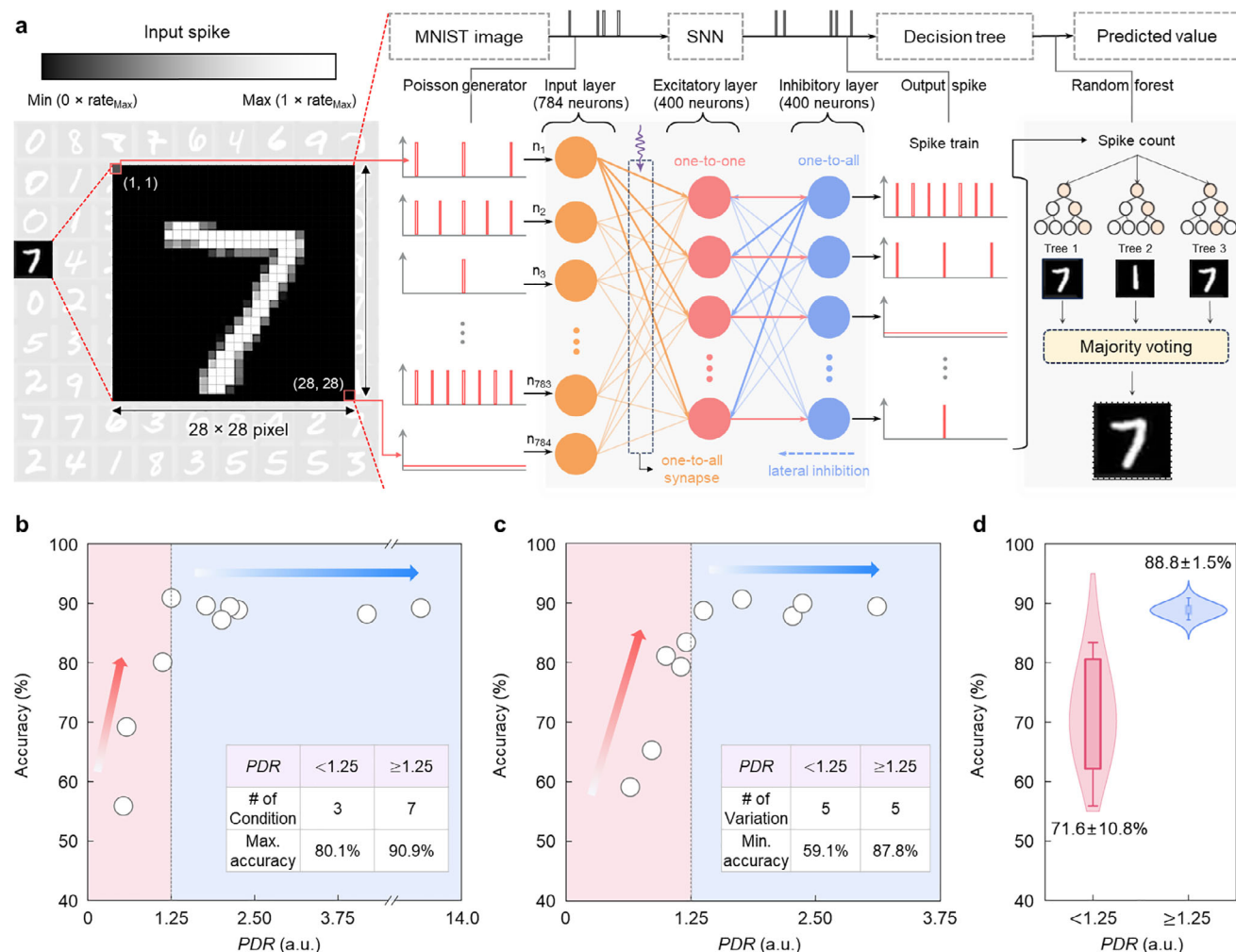


Figure 6. SNN simulation built with experimentally extracted STDP parameters from $\text{MoS}_2/\text{h-BN}$ heterojunction synaptic devices. a) Schematic of the network comprising 784 input, 400 excitatory, and 400 inhibitory neurons, with one-to-one excitatory–inhibitory pairing and one-to-all lateral inhibition for WTA competition. b, c) Maximum training accuracy of the SNN as a function of PDR, where the network was designed using STDP parameters from b) experimental (Conditions #1–#3) and c) modified (Variations #1–#10) STDP fitting curves. d) Distribution of the maximum training accuracies for the two PDR groups ($PDR < 1.25$, $PDR \geq 1.25$).

PDR rose from 1.1 to 2.3, peaked at 4.2, and then decreased to 2.1, demonstrating that precise adjustment of the $V_{\text{Pre,G}}$ amplitude enables fine control over the STDP symmetry. Figure 5h illustrates the dependence of STDP asymmetry on the I_{Light} amplitude. As the UV LED current increased from 0.1 to 0.3 A, both the LTP and LTD areas exhibited a consistent increase. As discussed in Condition #3, the higher UV intensity leads to more effective ionization of intrinsic defects within the h-BN layer, facilitating greater conductance modulation. Consequently, the PDR values steadily increased from 0.6 to 1.3 and finally reached 2.0, indicating that precise control over external optical stimulation can effectively tune the symmetry of the STDP curves. These results collectively demonstrate that the STDP characteristics of synaptic devices can be precisely tuned by carefully controlling the external parameters, enabling the hardware-level emulation of various biological synaptic behaviors ranging from the balanced plasticity observed in the hippocampus to the asymmetric STDP characteristics found in the neocortex.^[49,65]

STDP, the central learning rule in SNNs, modulates synaptic weights according to the timing between the presynaptic and postsynaptic spikes, enabling the extraction of salient input features without the need for labeled data. Because computation is invoked only when spikes occur, data movement and arithmetic operations are minimized, yielding ultra-low-power, and event-driven processing.^[9,66] To investigate the impact of different STDP characteristics on network performance, we conducted an SNN simulation using experimentally extracted STDP curves and their corresponding fitting parameters (A_+ , A_- , τ_+ , and τ_-), as summarized in Tables S1 and S2 (Supporting Information). Figure 6a shows a schematic of the SNN with 784 input neurons fully connected to 400 excitatory neurons; each excitatory neuron is paired one-to-one with an inhibitory neuron, which in turn connects one-to-all to excitatory neurons, enabling winner-takes-all competition via lateral inhibition.^[67] In the SNN, the synapses reflect the hardware STDP characteristics of A_+ , A_- , τ_+ , and τ_- , which were experimentally obtained from the PE synapses

under Conditions #1–#3 (Figure 5c–e; and Table S1, Supporting Information) and Variations #1–#10 (see Figure S16 and Table S2, Supporting Information). Based on these fitting parameters, the synaptic weights are dynamically updated according to the relative timing of the presynaptic and postsynaptic spikes. The network operates as follows. Each MNIST image with a size of 28×28 pixels is flattened into a 784×1 input vector and then converted into spike trains through a Poisson generator. The pixel intensity determines the input spiking rate: white pixels generate spikes at the maximum rate (rate_{Max}), while black pixels produce spikes at the minimum rate ($0 \times \text{rate}_{\text{Max}}$). Here, the rate_{Max} was set to 63.75 Hz to balance sufficient STDP-driven activation and minimal noise-induced firing. These spike trains are transmitted through the STDP-modulated synapses to the excitatory layer, where the timing-dependent learning mechanism adjusts the synaptic strength. The network employs one-to-one and one-to-all synapse configurations to implement lateral inhibition and competitive learning, allowing the most active excitatory neurons to inhibit other less active ones. For instance, when a particular excitatory neuron is strongly activated by a specific input pattern (e.g., an image of the digit “7”), it activates its paired inhibitory neuron, which then sends inhibitory signals to all the excitatory neurons. This feedback mechanism suppresses the firing of less active excitatory neurons by lowering their membrane potential, enhancing competition and selectivity. As a result, only specific excitatory neurons respond to input patterns, leading to more precise classification. The output spike train from the excitatory neurons serves as the network’s response. Each excitatory neuron counts the number of output spikes generated by the corresponding neuron during a given time window (i.e., 400 ms). Then, the number of spikes is used as the input features for a Random Forest classifier, which consists of multiple decision trees analyzing the spike patterns to determine the predicted digit. The final classification result is obtained through majority voting across the decision trees. This technique improves accuracy and ensures consistent performance across varying input patterns.

Figure 6b shows the maximum training accuracy of the SNN when the fitted STDP parameters—extracted from ten experimental STDP curves (Conditions #1–#3)—were applied. Each data point represents the maximum training accuracy against the LTP/D area ratio (PDR). For $PDR < 1.25$, the accuracy remained below 85% (the maximum was 80.1%) whereas the network consistently achieved over 85% (up to 90.9%) for $PDR \geq 1.25$. Here, $PDR = 1.25$ can be regarded as a saturation point for the designed SNN, corresponding to an LTP area 25% larger than the LTD area. At this point, potentiation sufficiently outweighs depression, allowing the synaptic weights to be maintained even under lateral inhibition competition, thereby enabling selective pattern activation. In contrast, when the PDR falls below this value, the potentiation might be too weak, and the synaptic weights gradually decay, suppressing neuron firing and degrading overall network performance. To further examine the significance of the PDR value, an STDP curve with a $PDR = 1.25$ (Condition #3, 0.2 A) was taken as the baseline, and its LTP and/or LTD curves were vertically shifted to create variations with ranging PDR values from 0.64 to 3.1 (see Table S3, Supporting Information). Figure 6c presents the maximum training accuracy obtained using each of the modified STDP curves (Variations #1–#10). When the PDR

exceeds 1.25, accuracy above 85% (the minimum was 87.8%) is consistently achieved. For PDR values below 1.25, the accuracy gradually increases with increasing PDR , showing a clear upward tendency. This result is consistent with the trend observed in Figure 6b and supports our assumption that a PDR of 1.25 represents the saturation point of the designed SNN. Figure 6d depicts the distribution of the maximum training accuracies for the two PDR groups ($PDR < 1.25$, $PDR \geq 1.25$). For $PDR \geq 1.25$ ($n = 12$), the mean accuracy was $88.8 \pm 1.5\%$, demonstrating both high performance and low variability. In contrast, for $PDR < 1.25$ ($n = 8$), the mean accuracy fell to $71.6 \pm 10.8\%$ (see Figure S17, Supporting Information). This disparity likely arises from the network’s strong lateral inhibition mechanism: when the PDR is low, LTD dominates over LTP, causing synaptic weights to weaken, neurons to lose the firing competition, and overall learning performance to decline. Conversely, when LTP exceeds LTD by at least 25% ($PDR \geq 1.25$), the synaptic weights withstand competitive suppression, enabling selective activation of specific neurons and thereby maintaining stable classification accuracy. Notably, the maximum training accuracy achieved in this work reached 90.9%, which is remarkable in view of the network’s structural simplicity and its reliance on unsupervised learning (Table 3).

3. Conclusion

In this work, we implemented a hardware-based multi-channel SNN that integrates $\text{MoS}_2/\text{h-BN}$ PE synaptic devices with analog LIF neuron circuits for the first time. This system leverages the precise timing dependency between the electrical presynaptic and optical postsynaptic spikes—an essential feature of STDP—to enable self-learning functionality. When electrical and optical pulses overlap within a specific timing window, UV illumination induces ionization or neutralization of the intrinsic defects in h-BN , resulting in the generation of charge carriers. These photogenerated carriers migrate into the MoS_2 channel depending on the polarity of the electrical pulse, leading to long-term potentiation (LTP) or long-term depression (LTD). The shape modulation of the presynaptic spike applied to the gate terminal of the synaptic device enabled emulation of the diverse biological STDP learning rules including Hebbian, anti-Hebbian, all-LTP, and all-LTD behaviors. Utilizing these rules, single- and multi-synapse SNNs were constructed, and their self-learning characteristics were demonstrated at the system level. Subsequently, the LTP/D area ratio (PDR) in the STDP characteristic curves for the synaptic device was further investigated by precisely controlling the operating conditions. Based on the extracted parameters of the STDP curves, the hardware SNN simulation was designed, and the system-level performance of the TD-SNNs was evaluated. With optimized STDP characteristics, the designed network achieved pattern classification accuracy up to 90.9%, confirming not only the functional validity but also the practical applicability of the proposed TD-SNN architecture. These collective results showed hardware-level timing-dependent learning from device to system scales. We believe that this concept will provide a fundamental basis for future adaptive and energy-efficient neuromorphic systems. We also note that wafer-level integration of 2D heterostructure devices, including the present photoelectroactive synapse, remains a challenging task. Such

Table 3. Performance comparison of the classification accuracy of our Brian2-based SNN model with previous studies on the MNIST dataset. While existing studies achieve accuracy in the range of 86.0%–90.0%, our study achieved 90.8% accuracy, showing competitive performance.

Device Type	Pixel	Input Neuron	Excitation/Inhibitory Layer	Accuracy	References
PE transistor ^{a)}	28 × 28	784	400/400	90.9%	Our work
Memristor	28 × 28	784	400/400	86.68%	[68]
Simulation	28 × 28	784	400/400	86.00%	[69]
Memristor	28 × 28	784	400/400	87.58%	[70]
Memristor	28 × 28	784	400/400	90.00%	[67]
SAF device ^{b)}	28 × 28	784	1600/1600	88.50%	[71]

^{a)} Photoelectroactive transistor. ^{b)} Synthetic anti-ferromagnetic device.

large-area growth can be achieved through chemical vapor deposition (CVD), which is regarded as a promising approach for the scalable fabrication of wafer-scale 2D materials. However, when 2D materials are grown using CVD, technical factors such as grain-boundary formation, defect density, stoichiometric variation, and contact optimization may arise.^[72–74] These factors may reduce the electrical or optical uniformity of the devices, so realizing reliable neuromorphic hardware demands clear understanding of 2D material nucleation and growth together with controlled interfacial interactions between reactants and substrates.

4. Experimental Section

Fabrication of the PE Synapse: The device was fabricated on a 90-nm-thick SiO₂/heavily boron-doped Si substrate. The *h*-BN and MoS₂ flakes were mechanically exfoliated and transferred using a dry transfer method. The presynaptic and postsynaptic terminals were patterned using electron-beam lithography, followed by deposition of 10 nm thick Ti and 80 nm thick Au to form the electrodes. The distance between the two electrodes was 4 μm, while the width of the electrodes was 5 μm.

Implementation of the LIF Neuron Circuit: The LIF neuron circuit was designed using OrCAD Capture CIS and PCB Editor (Cadence Design Systems, Inc., San Jose, CA, USA), and subsequently fabricated by integrating passive and active components, such as resistors (*R*), capacitors (*C*), inductors (*L*), diodes, operational amplifiers (OP-AMP), transistors, IC chips, and a microprocessor. The core functionalities—such as leaky integrate-and-fire—of the LIF neuron circuit were implemented using the following specific components: i) A pulse generator, in which a microprocessor (AD9106, Analog Devices, Inc., MA, USA) was used to supply the input voltage pulses, such as *V*_{Pre,D}, *V*_{Pre,C}, and *V*_{Post}; ii) The LIU, which was realized as an OP-AMP-based integrator (AD8652, Analog Devices, Inc., MA, USA) connected with *C*_{Mem} and *R*_{Leak}, where *C*_{Mem} is charged by *V*_{Mem} and continuously discharged through *R*_{Leak} to perform the leaky integrate operation; iii) The FU was implemented with a comparator (LTC6752, Analog Devices, Inc., MA, USA). The reference voltage of the comparator corresponds to the firing threshold voltage (*V*_{Thres}). When *V*_{Mem}, delivered from the LIU, exceeds *V*_{Thres}, the FU generates a *V*_{Fire}; iv) A 365 nm UV LED (Epistar, HOTRED, Shenzhen Qutai Photoelectric Co., Ltd., GD, CN) was employed to apply a *V*_{Post} signal to the synaptic device in response to the *V*_{Fire} generated from the FU. Upon the occurrence of *V*_{Fire}, the LED irradiated UV light onto the synaptic device, enabling conductance modulation in conjunction with *V*_{Pre,C}. (See Figures S9 and S18, Supporting Information).

Integration of PE Synapses with a LIF Neuron Circuit for TD-SNN Implementation: The TD-SNN consisted of PE synaptic devices and a leaky integrate-and-fire (LIF) neuron circuit. The PE synaptic device was wire-bonded to a 28-pin ceramic side-brazed dual in-line package (DIP) using

thermosonic bonding with a TPT HB05 Wedge & Ball Bonder (TPT Wire Bonder GmbH & Co., KG, Germany). This process established electrical connections between the synaptic devices and the LIF neuron circuit, enabling subsequent characterization and testing.

Characterization of the Synaptic Device and Neuron Circuit: The image of the synaptic device was obtained using an Olympus BX53M optical microscope. The electrical properties were characterized by utilizing a Keithley 4200A-SCS parameter analyzer, while the functionality of the neuron board was evaluated using a Keysight DSOX3014T oscilloscope. To induce the PE synaptic response of the device, an external UV LED with a wavelength of 365 nm was used to illuminate the device. Photoluminescence (PL) measurements were carried out under steady-state conditions at 70 K using a Horiba Jobin Yvon LabRAM HR Evolution system (Visible/NIR). A 325 nm argon laser, with a beam spot size of ≈1–2 μm, served as the excitation source, and the emitted signal was captured using a charge-coupled device (CCD) detector. Absorption measurements were conducted at room temperature employing a Shimadzu UV-3600 UV-Vis-NIR spectrophotometer, capable of covering a broad spectral range from 200 to 3000 nm.

Implementation of the Brian2 SNN Simulator: The SNN simulation was conducted using the Brian2 simulator in Python, employing a network architecture designed to reflect the measured STDP characteristics of the MoS₂/*h*-BN PE synaptic devices.^[67] The STDP curve was obtained from a single representative device (*N* = 1) and fitted using an exponential decay model (Equation (1)).

$$\Delta G = \begin{cases} A_+ e^{-\frac{\Delta t}{\tau_+}} & \text{for } \Delta t > 0 \\ -A_- e^{\frac{\Delta t}{\tau_-}} & \text{for } \Delta t < 0 \end{cases} \quad (1)$$

The extracted parameters (*A*₊, *A*_−, *τ*₊, and *τ*_−) were directly used as update coefficients in the simulation. Here, *A*₊ and *A*_− represent the amplitude coefficients corresponding to the maximum conductance change during LTP and LTD, respectively. *τ*₊ and *τ*_− denote the characteristic time constants that determine how rapidly the potentiation and depression effects decay with increasing |Δ*t*|. The network consisted of 784 input neurons, 400 excitatory neurons, and 400 inhibitory neurons. Each MNIST image (28 × 28 pixels) was encoded as an input spike train, with spike rates proportional to the pixel intensity, and presented to the network for 0.4 seconds per image. If the total spike count from all the excitatory neurons was less than five, the same image was represented with proportionally increased input rates (up to 8.5 times, maximum 15 trials) to ensure sufficient activity for STDP learning. After each presentation, a 0.4 s reset period was used to minimize the carryover effects between the images. Synaptic plasticity between the input and excitatory neurons was implemented as an event-driven process using Brian2's on_pre event and on_post mechanisms. At each presynaptic spike (on_pre event, LTD, Δ*t* < 0), the synaptic weight was updated according to Equation (2).

$$W_{new} = W_{current} - (\eta) \times A_{Post} \quad (2)$$

At each postsynaptic spike (on_post event, LTP, $\Delta t > 0$), the synaptic weight was updated according to Equation (3).

$$W_{new} = W_{current} + (\eta) \times A_{Pre} \quad (3)$$

where $W_{current}$ and W_{new} are the synaptic weights before and after each update, η is the learning rate (0.1 during training), and A_{Pre} and A_{Post} are the presynaptic and postsynaptic trace variables, which are incremented by A_+ and A_- at each presynaptic and postsynaptic spike, respectively, but otherwise decay exponentially with τ_+ and τ_- . After unsupervised training with 60000 MNIST images, the synaptic weights were fixed ($\eta = 0$) and the evaluation phase commenced. For each test image, the input spike trains were presented for 0.4 s, and the spike counts from all of the 400 excitatory neurons were recorded to form a 400-dimensional feature vector. This feature vector was classified using scikit-learn's RandomForestClassifier, and the predicted class was compared to the true label to calculate the accuracy.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

2D materials, multi-channel timing-dependent spiking neural networks, neuromorphic devices, photoelectroactive artificial synapses, van der Waals heterojunctions

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- [1] C. Mead, *Proc. IEEE* **1990**, *78*, 1629.
- [2] D. Ham, H. Park, S. Hwang, K. Kim, *Nat. Electron.* **2021**, *4*, 635.
- [3] S. Seo, J.-J. Lee, H.-J. Lee, H. W. Lee, S. Oh, J. J. Lee, K. Heo, J.-H. Park, *ACS Appl. Electron. Mater.* **2020**, *2*, 371.
- [4] Y. Andrade-Talavera, A. Fisahn, A. Rodríguez-Moreno, *Mol. Psychiatry* **2023**, *28*, 2177.
- [5] K. Roy, A. Jaiswal, P. Panda, *Nature* **2019**, *575*, 607.
- [6] M. Baronig, R. Ferrand, S. Sabathiel, R. Legenstein, *Nat. Commun.* **2025**, *16*, 5776.
- [7] M. Yao, O. Richter, G. Zhao, N. Qiao, Y. Xing, D. Wang, T. Hu, W. Fang, T. Demirci, M. De Marchi, L. Deng, T. Yan, C. Nielsen, S. Sheik, C. Wu, Y. Tian, B. Xu, G. Li, *Nat. Commun.* **2024**, *15*, 4464.
- [8] Z. Wang, H. Wu, G. W. Burr, C. S. Hwang, K. L. Wang, Q. Xia, J. J. Yang, *Nat. Rev. Mater.* **2020**, *5*, 173.
- [9] Z. Yi, J. Lian, Q. Liu, H. Zhu, D. Liang, J. Liu, *Neurocomputing* **2023**, *531*, 163.
- [10] A. Siddique, M. I. Vai, S. H. Pun, *Sci. Rep.* **2023**, *13*, 6280.
- [11] S. K. Das, S. K. Nandi, C. V. Marquez, A. Rúa, M. Uenuma, S. K. Nath, S. Zhang, C.-H. Lin, D. Chu, T. Ratcliff, R. G. Elliman, *Adv. Intell. Syst.* **2024**, *6*, 2400191.
- [12] J. H. Lee, T. Delbruck, M. Pfeiffer, *Front. Neurosci.* **2016**, *10*, 508.
- [13] C. Lee, S. S. Sarwar, P. Panda, G. Srinivasan, K. Roy, *Front. Neurosci.* **2020**, *14*, 119.
- [14] B. Rueckauer, I.-A. Lungu, Y. Hu, M. Pfeiffer, S.-C. Liu, *Front. Neurosci.* **2017**, *11*, 682.
- [15] M. A. Zidan, J. P. Strachan, W. D. Lu, *Nat. Electron.* **2018**, *1*, 22.
- [16] N. A. Rahman, N. Yusoff, *Neurocomputing* **2025**, *618*, 129170.
- [17] Y. Jo, D. Y. Woo, G. Noh, E. Park, M. J. Kim, Y. W. Sung, D. K. Lee, J. Park, J. Kim, Y. Jeong, S. Lee, I. Kim, J.-K. Park, S. Park, J. Y. Kwak, *Adv. Funct. Mater.* **2024**, *34*, 2309058.
- [18] M. Huang, L. Xu, J. A. del Alamo, J. Li, B. Yildiz, *Adv. Mater.* **2025**, *37*, 2418484.
- [19] M. V. DeBole, B. Taba, A. Amir, F. Akopyan, A. Andreopoulos, W. P. Risk, J. Kusnitz, C. Ortega Otero, T. K. Nayak, R. Appuswamy, P. J. Carlson, A. S. Cassidy, P. Datta, S. K. Esser, G. J. Garreau, K. L. Holland, S. Lekuch, M. Mastro, J. McKinstry, C. di Nolfo, B. Paulovicks, J. Sawada, K. Schleupen, B. G. Shaw, J. L. Klamro, M. D. Flickner, J. V. Arthur, D. S. Modha, *Computer* **2019**, *52*, 20.
- [20] M. Davies, A. Wild, G. Orchard, Y. Sandamirskaya, G. A. F. Guerra, P. Joshi, P. Plank, S. R. Risbud, *Proc. IEEE* **2021**, *109*, 911.
- [21] J. Pei, L. Deng, S. Song, M. Zhao, Y. Zhang, S. Wu, G. Wang, Z. Zou, Z. Wu, W. He, F. Chen, N. Deng, S. Wu, Y. Wang, Y. Wu, Z. Yang, C. Ma, G. Li, W. Han, H. Li, H. Wu, R. Zhao, Y. Xie, L. Shi, *Nature* **2019**, *572*, 106.
- [22] S. Pazos, K. Zhu, M. A. Villena, O. Alharbi, W. Zheng, Y. Shen, Y. Yuan, Y. Ping, M. Lanza, *Nature* **2025**, *640*, 69.
- [23] H. Xu, D. Shang, Q. Luo, J. An, Y. Li, S. Wu, Z. Yao, W. Zhang, X. Xu, C. Dou, H. Jiang, L. Pan, X. Zhang, M. Wang, Z. Wang, J. Tang, Q. Liu, M. Liu, *Nat. Commun.* **2023**, *14*, 6385.
- [24] K.-H. Lee, D. Kwon, I.-S. Lee, J. Hwang, J. Im, J.-H. Bae, W. Y. Choi, S. Y. Woo, J.-H. Lee, *Adv. Intell. Syst.* **2024**, *6*, 2300490.
- [25] J. Park, M.-W. Kwon, H. Kim, S. Hwang, J.-J. Lee, B.-G. Park, *IEEE Trans. Electron Devices* **2017**, *64*, 2438.
- [26] I.-J. Kim, J.-S. Lee, *Sci. Adv.* **2024**, *10*, 1345eadn.
- [27] Y. Choi, S. Oh, C. Qian, J.-H. Park, J. H. Cho, *Nat. Commun.* **2020**, *11*, 4595.
- [28] G. Lee, Y. E. Kim, H. Kim, H.-K. Lee, J. Y. Park, S. Oh, H. Yoo, *Small* **2025**, *21*, 2501997.
- [29] J. Park, Y. Yun, S. Bae, Y. Jang, S. Shin, S.-Y. Lee, *Adv. Sci.* **2025**, *12*, 2500568.
- [30] S. Oh, J.-I. Cho, B. H. Lee, S. Seo, J.-H. Lee, H. Choo, K. Heo, S. Y. Lee, J.-H. Park, *Sci. Adv.* **2021**, *7*, abg9450.
- [31] H. Ahn, Y. Kim, S. Seo, J. Lee, S. Lee, S. Oh, B. Kim, J. Park, S. Kang, Y. Kim, A. Ham, J. Lee, D. Park, S. Kwon, D. Lee, J.-E. Ryu, J.-C. Shin,

- A. Sahasrabudhe, K. S. Kim, S.-H. Bae, K. Kang, J. Kim, S. Oh, J.-H. Park, *Adv. Mater.* **2025**, *37*, 2418582.
- [32] S. Seo, J.-J. Lee, R.-G. Lee, T. H. Kim, S. Park, S. Jung, H.-K. Lee, M. Andreev, K.-B. Lee, K.-S. Jung, S. Oh, H.-J. Lee, K. S. Kim, G. Y. Yeom, Y.-H. Kim, J.-H. Park, *Adv. Mater.* **2021**, *33*, 2102980.
- [33] S. Seo, B.-S. Kang, J.-J. Lee, H.-J. Ryu, S. Kim, H. Kim, S. Oh, J. Shim, K. Heo, S. Oh, J.-H. Park, *Nat. Commun.* **2020**, *11*, 3936.
- [34] Y. Jeong, H. J. Lee, J. Park, S. Lee, H.-J. Jin, S. Park, H. Cho, S. Hong, T. Kim, K. Kim, S. Choi, S. Im, *npj 2D Mater. Appl.* **2022**, *6*, 23.
- [35] S. Park, Y. Jeong, H.-J. Jin, J. Park, H. Jang, S. Lee, W. Huh, H. Cho, H. G. Shin, K. Kim, C.-H. Lee, S. Choi, S. Im, *ACS Nano* **2020**, *14*, 12064.
- [36] F. Zhang, C. Li, Z. Li, L. Dong, J. Zhao, *Microsyst. Nanoeng.* **2023**, *9*, 16.
- [37] W. Wang, N. Clark, M. Hamer, A. Carl, E. Tovari, S. Sullivan-Allsop, E. Tillotson, Y. Gao, H. de Latour, F. Selles, J. Howarth, E. G. Castanon, M. Zhou, H. Bai, X. Li, A. Weston, K. Watanabe, T. Taniguchi, C. Mattevi, T. H. Bointon, P. V. Wiper, A. J. Strudwick, L. A. Ponomarenko, A. V. Kretinin, S. J. Haigh, A. Summerfield, R. Gorbachev, *Nat. Electron.* **2023**, *6*, 981.
- [38] B. Karmakar, B. Das, S. Mandal, R. Paramanik, S. Maity, T. Kundu, S. Das, M. Palit, K. Dey, K. Dolui, S. Datta, *Phys. Rev. Appl.* **2025**, *23*, 054013.
- [39] C. He, J. Tang, D.-S. Shang, J. Tang, Y. Xi, S. Wang, N. Li, Q. Zhang, J.-K. Lu, Z. Wei, Q. Wang, C. Shen, J. Li, S. Shen, J. Shen, R. Yang, D. Shi, H. Wu, S. Wang, G. Zhang, *ACS Appl. Mater. Interfaces* **2020**, *12*, 11945.
- [40] H. Tian, X. Cao, Y. Xie, X. Yan, A. Kostelec, D. DiMarzio, C. Chang, L.-D. Zhao, W. Wu, J. Tice, J. J. Cha, J. Guo, H. Wang, *ACS Nano* **2017**, *11*, 7156.
- [41] Y. Choi, S. Jeong, H. Jeong, S. Han, J. Ko, S. E. Yu, Z. Xu, M. S. Chae, M. Son, Y. Meng, S. Xu, J.-H. Kang, S. Mun, S.-H. Bae, *npj Unconv. Comput.* **2025**, *2*, 8.
- [42] G. Bi, M. Poo, *J. Neurosci.* **1998**, *18*, 10464.
- [43] S. Oh, J.-J. Lee, S. Seo, G. Yoo, J.-H. Park, *npj 2D Mater. Appl.* **2021**, *5*, 95.
- [44] P. Desai-Chowdhry, A. B. Brummer, V. M. Savage, *Sci. Rep.* **2022**, *12*, 20810.
- [45] W. C. Abraham, O. D. Jones, D. L. Glanzman, *npj Sci. Learn.* **2019**, *4*, 9.
- [46] N. Caporale, Y. Dan, *Annu. Rev. Neurosci.* **2008**, *31*, 25.
- [47] C. Piochon, P. Kruskal, J. N. MacLean, C. Hansel, *Front. Neural Circuits* **2013**, *6*, 124.
- [48] E. Pali, S. Masoli, D. Di Domenico, T. Sorbo, F. Prestori, E. D'Angelo, *Commun. Biol.* **2025**, *8*, 731.
- [49] R. K. Mishra, S. Kim, S. J. Guzman, P. Jonas, *Nat. Commun.* **2016**, *7*, 11552.
- [50] H. Salgado, G. Köhr, M. Treviño, *Sci. Rep.* **2012**, *2*, 417.
- [51] G. H. Seol, J. Ziburkus, S. Huang, L. Song, I. T. Kim, K. Takamiya, R. L. Huganir, H.-K. Lee, A. Kirkwood, *Neuron* **2007**, *55*, 919.
- [52] P. Smolen, D. A. Baxter, J. H. Byrne, *Front. Comput. Neurosci.* **2020**, *14*, 569349.
- [53] K. Pozo, Y. Goda, *Neuron* **2010**, *66*, 337.
- [54] G. Dellaferrera, S. Woźniak, G. Indiveri, A. Pantazi, E. Eleftheriou, *Nat. Commun.* **2022**, *13*, 1885.
- [55] J. Kim, E. C. Park, W. Shin, R.-H. Koo, J. Im, C.-H. Han, J.-H. Lee, D. Kwon, *Adv. Sci.* **2024**, *11*, 2407870.
- [56] H. Choi, S. Baek, H. Jung, T. Kang, S. Lee, J. Jeon, B. C. Jang, S. Lee, *Adv. Mater.* **2025**, *37*, 2406970.
- [57] Y. Wang, Y. Zhang, Y. Wang, X. Wang, H. Zhang, R. Xu, Y. Tong, *Adv. Electron. Mater.* **2025**, *11*, 2400421.
- [58] M. Park, J. Y. Yang, M. J. Yeom, B. Bae, Y. Baek, G. Yoo, K. Lee, *Sci. Adv.* **2023**, *9*, adh9889.
- [59] R. A. John, J. Acharya, C. Zhu, A. Surendran, S. K. Bose, A. Chaturvedi, N. Tiwari, Y. Gao, Y. He, K. K. Zhang, M. Xu, W. L. Leong, Z. Liu, A. Basu, N. Mathews, *Nat. Commun.* **2020**, *11*, 3211.
- [60] S.-I. Amari, M. A. Arbib, *Competition and Cooperation in Neural Nets*, Springer, Berlin **1977**, 119.
- [61] S. Eckmann, E. J. Young, J. Gjorgjieva, *Proc. Natl. Acad. Sci. USA* **2024**, *121*, 2305326121.
- [62] Z. Brzosko, S. B. Mierau, O. Paulsen, *Neuron* **2019**, *103*, 563.
- [63] G. Malavena, A. S. Spinelli, C. M. Compagnoni, presented at **2018 IEEE International Electron Devices Meeting (IEDM)**, San Francisco, CA, USA, December **2018**.
- [64] C. Casellato, A. Antonietti, J. A. Garrido, G. Ferrigno, E. D'Angelo, A. Pedrocchi, *Front. Comput. Neurosci.* **2015**, *9*, 24.
- [65] P. J. Sjöström, G. G. Turrigiano, S. B. Nelson, *Neuron* **2001**, *32*, 1149.
- [66] M. Pfeiffer, T. Pfeil, *Front. Neurosci.* **2018**, *12*, 774.
- [67] K. Zhu, S. Pazos, F. Aguirre, Y. Shen, Y. Yuan, W. Zheng, O. Alharbi, M. A. Villena, B. Fang, X. Li, A. Milozzi, M. Farronato, M. Muñoz-Rojo, T. Wang, R. Li, H. Fariborzi, J. B. Roldan, G. Benstetter, X. Zhang, H. N. Alshareef, T. Grasser, H. Wu, D. Ielmini, M. Lanza, *Nature* **2023**, *618*, 57.
- [68] J. Feng, Y. Fan, Y. Wang, Q. Song, Y. Liu, Y. Chen, D. Li, W. Huang, *Adv. Funct. Mater.* **2025**, *35*, 2420547.
- [69] R. Naderi, A. Rezaei, M. Amiri, H. Peremans, *Sci. Rep.* **2025**, *15*, 17647.
- [70] S. Yang, D. Li, J. Feng, B. Gong, Q. Song, Y. Wang, Z. Yang, Y. Chen, Q. Chen, W. Huang, *Adv. Electron. Mater.* **2024**, *10*, 2400075.
- [71] D. Wang, R. Tang, H. Lin, L. Liu, N. Xu, Y. Sun, X. Zhao, Z. Wang, D. Wang, Z. Mai, Y. Zhou, N. Gao, C. Song, L. Zhu, T. Wu, M. Liu, G. Xing, *Nat. Commun.* **2023**, *14*, 1068.
- [72] L. Tang, J. Tan, H. Nong, B. Liu, H.-M. Cheng, *Acc. Mater. Res.* **2021**, *2*, 36.
- [73] S. Bhowmik, A. Govind Rajan, *iScience* **2022**, *25*, 103832.
- [74] Z. Han, L. Li, F. Jiao, G. Yu, Z. Wei, D. Geng, W. Hu, *Nanoscale Adv.* **2021**, *3*, 6545.